

GradTDDFT: A Differentiable Time-Dependent Density Functional Theory Framework for Training Neural Network Exchange-Correlation Functionals

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GradTDDFT is a JAX-based framework for differentiable time-dependent density functional theory (TDDFT) with neural exchange-correlation (XC) functionals. The method separates a fixed molecular data layer—basis functions, quadrature grids, one-electron integrals, and electron-repulsion integrals—from the differentiable Kohn–Sham SCF and TDDFT response layers. This separation gives a well-defined computational graph from neural XC parameters to self-consistent orbitals, TDA/Casida response matrices, excitation energies, and oscillator strengths. The central technical element is an energy-consistent local response tensor: the XC potential and f_{xc} kernel are obtained as first and second derivatives of the same learned local energy density by JAX automatic differentiation. The neural functional uses a coefficient-basis architecture in which pointwise density, gradient, kinetic-energy-density, local exact-exchange, and optional MP2/PT2 descriptors produce mixing fields over fixed semilocal, HF, and MP2/PT2 channels. We formulate the resulting ground-state and excited-state objectives together with the corresponding computational evaluation protocol.

Keywords: differentiable programming, JAX, TDDFT, neural exchange-correlation functional, excited states

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I. INTRODUCTION

Time-dependent density functional theory (TDDFT) within the linear-response formalism of Casida¹ is one of the standard approaches for computing molecular excitation energies and oscillator strengths. Its cost is often compatible with chemical screening studies, but its accuracy is controlled by the ground-state exchange-correlation (XC) functional and by the associated response kernel. Conventional semilocal and hybrid functionals therefore inherit known limitations for charge-transfer excitations, Rydberg states, static correlation, and double-excitation character. These failures are not merely post-processing errors: the same density functional determines the SCF density, orbital energy gaps, the XC potential, and the kernel entering the TDDFT response matrices.

Neural network XC functionals offer a route to learn more flexible functional forms, including nonlocal XC structure, from accurate reference data.^{2,3} For excited states, however, fitting ground-state energies alone is not enough: a functional used in TDDFT must also produce stable self-consistent densities, useful frontier orbital gaps, and a response kernel consistent with the learned XC energy. A recent overview of differentiable DFT emphasizes that this changes the role of the KS equations: the functional is not only evaluated inside a self-consistent DFT calculation, but trained through the differentiable SCF map, with gradients from observables propagated back to the model parameters while the governing electronic-structure equations remain in the loop.⁴ For TDDFT, the corresponding differentiable program must extend beyond the ground-state SCF fixed point to include response-kernel construction, TDA/Casida eigenvalue problems, excitation energies, oscillator strengths, and spectrum-level objectives. Differentiable training is therefore important because it can place losses on the quantities actually used to judge excited-state predictions, rather than only on pre-response ground-state energies. This differentiable view also changes how the XC derivatives are implemented. In conventional DFT codes, the XC potential $v_{xc} = \delta E_{xc}/\delta\rho$ and the TDDFT kernel $f_{xc} = \delta^2 E_{xc}/\delta\rho\delta\rho'$ are usually derived analytically for each functional form and then encoded as separate routines. GradTDDFT instead treats the local XC energy density as the primary differentiable object and obtains these first- and second-order derivatives by automatic differentiation, so the SCF potential and the response kernel remain tied to the same energy expression.⁵

This idea is part of a broader movement toward differentiable scientific codes. At the level

of methods and software, D4FT and DQC expose differentiable DFT workflows,^{6,7} PySCFAD differentiates PySCF-style mean-field and post-mean-field calculations,⁸ GradDFT focuses on machine-learning enhanced ground-state density functionals,⁹ and jax-xc translates conventional Libxc functional forms into JAX-compatible energy-density routines.¹⁰ Differentiable semiempirical models provide a complementary low-cost route: dxtb implements differentiable extended tight-binding, and related semiempirical formulations have reported ground- and excited-state gradients.^{11,12} At the level of molecular and materials applications, JAX-MD provides differentiable simulation primitives for molecular systems,¹³ JAX-based DFT workflows have been developed for crystalline systems and direct solid-state optimization,^{14,15} plane-wave AD-DFPT has been applied to materials design, error control, and parameter learning,¹⁶ and orbital-free DFT has used automatic differentiation to obtain functional potentials and higher-order material properties directly from energy expressions.¹⁷ These developments demonstrate that automatic differentiation can turn molecular simulation and electronic-structure calculations into trainable numerical programs. The missing piece for excited states is an end-to-end differentiable TDDFT formulation in which the neural XC energy, SCF potential, response kernel, and excited-state observables belong to the same differentiable computational graph.

We present GradTDDFT to address this gap. The framework keeps PySCF as the source of molecular construction, basis information, grids, and integrals, while expressing the SCF, XC, TDDFT response, and training logic in JAX. GradTDDFT separates fixed molecular data from the differentiable KS-SCF and linear-response layers, giving a continuous map from neural XC parameters to self-consistent densities, orbital rotations, TDA/Casida matrices, excitation energies, oscillator strengths, and excited-state objectives. The response eigenproblem is solved by a Davidson iteration in the forward pass and differentiated using the implicit eigensolver strategy of Lei Wang and co-workers,¹⁸ so gradients are reconstructed from the converged Ritz pairs rather than from the full iterative solver history. The neural functional is written as a coefficient-basis family in which local density descriptors generate mixing fields over semilocal, HF, and optional MP2/PT2 channels. For semilocal channels, the XC energy, potential, and local f_{xc} tensor are obtained from the same energy-density expression. The HF channel supplies a scalar hybrid exchange response, while the MP2/PT2 channel supplies a root-specific second-order correction after the TDA/Casida singles problem is solved. This design provides separate ground-state and excited-state training protocols

within the same differentiable implementation.

II. THEORY AND METHODS

The main theoretical construction follows the organization used in GradDFT-style differentiable functional work⁹: first define the energy functional and its parameterization, then specify the response equations used to obtain observables, and finally state how losses are differentiated through the SCF and response maps. Detailed descriptor definitions, response-kernel contractions, and neural hybrid channel definitions are deferred to the Supporting Information.

A. DFT and Neural Hybrid XC Functional

For a molecular system with density ρ , the Kohn–Sham energy is written as

$$E_\theta[\rho] = T_s[\rho] + E_{\text{ne}}[\rho] + J[\rho] + E_{\text{xc}}^\theta[\rho] + E_{\text{nn}}, \quad (1)$$

where T_s is the noninteracting kinetic energy, E_{ne} is the electron–nuclear attraction, J is the Hartree energy, E_{nn} is the nuclear repulsion, and θ denotes the parameters of the learned exchange–correlation (XC) model. The corresponding generalized eigenvalue problem in an AO basis is

$$\mathbf{F}_\theta[\mathbf{D}]\mathbf{C} = \mathbf{S}\mathbf{C}\epsilon, \quad \mathbf{F}_\theta = \mathbf{h} + \mathbf{J}[\mathbf{D}] + \mathbf{V}_{\text{xc}}^\theta[\mathbf{D}] + \mathbf{V}_{\text{hyb}}^\theta[\mathbf{D}], \quad (2)$$

where $\mathbf{D}$ is the density matrix and $\mathbf{V}_{\text{hyb}}^\theta$ collects nonlocal hybrid contributions.

The neural functional is expressed as a coefficient-basis hybrid functional with three physically motivated channel groups: local/semi-local XC channels, a local exact-exchange HF channel, and a local MP2/PT2 second-order correlation channel. This mirrors the doubly local double-hybrid idea, where local mixing functions control both exact exchange and PT2 correlation in real space.¹⁹ On a quadrature grid,

$$E_{\text{xc}}^\theta = \sum_g w_g e_{\text{xc}}^\theta(g), \quad e_{\text{xc}}^\theta(g) = \sum_{m=1}^M c_m^\theta(\mathbf{z}_g) e_m^{\text{sl}}(g) + c_x^\theta(\mathbf{z}_g) h_g + c_2^\theta(\mathbf{z}_g) p_g. \quad (3)$$

Here $\mathbf{z}_g$ is a local descriptor vector, c_m^θ , c_x^θ , and c_2^θ are learned mixing fields, and e_m^{sl} are local or semi-local functional energy densities. A general semi-local channel has the form

$$e_m^{\text{sl}}(\mathbf{r}) = \rho(\mathbf{r}) \varepsilon_m^{\text{sl}}(\rho_\sigma, \nabla \rho_\sigma \cdot \nabla \rho_{\sigma'}, \tau_\sigma), \quad (4)$$

with LDA, GGA, and meta-GGA channels obtained by selecting the corresponding subset of density variables. For example, a spin-unpolarized LDA exchange channel and a GGA exchange channel may be written as

$$e_x^{\text{LDA}}(\mathbf{r}) = -C_x \rho(\mathbf{r})^{4/3}, \quad e_x^{\text{GGA}}(\mathbf{r}) = e_x^{\text{LDA}}(\mathbf{r}) F_x(s), \quad s = \frac{|\nabla \rho(\mathbf{r})|}{2(3\pi^2)^{1/3} \rho(\mathbf{r})^{4/3}}, \quad (5)$$

where $C_x = \frac{3}{4}(3/\pi)^{1/3}$ and F_x is the functional-specific GGA enhancement factor. Through jax-xc¹⁰, different conventional semi-local functional families can be inserted into this basis without changing the SCF or TDDFT equations.

The HF and MP2/PT2 ingredients are exposed to the neural model only through local channel densities,

$$\begin{aligned} E_x^{\text{HF}} &= \int h(\mathbf{r}) d\mathbf{r} \simeq \sum_g w_g h_g, \\ E_c^{(2)} &= \int p(\mathbf{r}) d\mathbf{r} \simeq \sum_g w_g p_g. \end{aligned} \quad (6)$$

The corresponding grid descriptors are

$$\begin{aligned} h_g &= -\frac{1}{2} \sum_{\sigma} \sum_{ij}^{\text{occ}} n_{i\sigma} n_{j\sigma} \phi_{i\sigma}(\mathbf{r}_g) \phi_{j\sigma}(\mathbf{r}_g) v_{ij\sigma}^x(\mathbf{r}_g), \\ p_g &= \frac{1}{4} \sum_{ijab} t_{ij}^{ab} \eta_{ij}^{ab}(\mathbf{r}_g). \end{aligned} \quad (7)$$

Thus $c_x^{\theta}(\mathbf{z}_g)$ and $c_2^{\theta}(\mathbf{z}_g)$ are pointwise mixing fields for the local exact-exchange and second-order correlation channels, rather than global scalar fractions. The orbital-pair potential $v_{ij\sigma}^x$, the second-order amplitude t_{ij}^{ab} , and the real-space pair projection η_{ij}^{ab} are defined in the Supporting Information. Constant coefficients in Eq. 3 recover a conventional hybrid or double-hybrid form, while learned pointwise coefficients define the neural hybrid functional used in GradTDDFT.

B. Linear-Response TDDFT, TDA, and Casida Equations

After SCF convergence, excitation energies are obtained from linear-response TDDFT. For occupied orbitals i, j and virtual orbitals a, b , the full Casida equation is

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix} \begin{pmatrix} \mathbf{X}_I \\ \mathbf{Y}_I \end{pmatrix} = \Omega_I \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & -\mathbf{1} \end{pmatrix} \begin{pmatrix} \mathbf{X}_I \\ \mathbf{Y}_I \end{pmatrix}, \quad (8)$$

where Ω_I is the excitation energy. The response matrices have the schematic form

$$A_{ia,jb} = (\epsilon_a - \epsilon_i)\delta_{ij}\delta_{ab} + K_{ia,jb}, \quad (9)$$

$$B_{ia,jb} = K_{ia,bj}, \quad (10)$$

where K contains Coulomb, XC-kernel, and hybrid-response contributions. For the semi-local part,

$$f_{xc}^\theta(\mathbf{r}, \mathbf{r}') = \frac{\delta^2 E_{xc}^\theta}{\delta\rho(\mathbf{r})\delta\rho(\mathbf{r}')}, \quad (11)$$

so the potential used in SCF and the local kernel used in TDDFT are derivatives of the same energy-density model. HF and MP2/PT2 channels are assigned distinct response constructions in the excited-state calculations. For the HF channel, the pointwise coefficient field is reduced to a density-weighted scalar exchange fraction,

$$a_x^\theta = \frac{\sum_g w_g \rho_g c_x^\theta(\mathbf{z}_g)}{\sum_g w_g \rho_g}, \quad (12)$$

and the TDDFT matrices use the conventional hybrid exact-exchange response

$$K_{ia,jb}^{x,A} = -a_x^\theta(ij|ab), \quad K_{ia,jb}^{x,B} = -a_x^\theta(ib|aj). \quad (13)$$

The local HF descriptor h_g can still condition the learned coefficients, but it does not introduce a separate orbital-response kernel for the projected local HF field.

For the PT2 channel, GradTDDFT follows the double-hybrid excited-state logic of solving the singles response first and then adding a root-specific second-order doubles correction.^{20,21} The learned PT2 coefficient field defines

$$a_c^\theta = \frac{\sum_g w_g \rho_g c_2^\theta(\mathbf{z}_g)}{\sum_g w_g \rho_g}, \quad 0 \leq a_c^\theta \leq 1, \quad (14)$$

and the final excitation energy is evaluated as

$$\Omega_I = \Omega_I^{\text{LR}} + a_c^\theta \Delta_I^{\text{CIS(D)}}. \quad (15)$$

Here Ω_I^{LR} is the TDA or Casida root obtained from the semilocal and HF response matrices, and $\Delta_I^{\text{CIS(D)}}$ is the state-specific second-order correction computed from the singles amplitudes and the occupied–virtual energy denominators. Thus the PT2 channel enters excited-state energies as a doubles correction rather than as an additional local response tensor in the TDDFT matrix.

The Tamm–Dancoff form sets $\mathbf{B} = 0$, yielding the Hermitian eigenvalue problem

$$\mathbf{A}\mathbf{X}_I = \Omega_I\mathbf{X}_I. \quad (16)$$

Oscillator strengths are evaluated from the transition dipole,

$$\boldsymbol{\mu}_I = \sum_{ia} (X_{ia}^I + Y_{ia}^I) \boldsymbol{\mu}_{ia}, \quad f_I = \frac{2}{3} \Omega_I |\boldsymbol{\mu}_I|^2, \quad (17)$$

with $Y_{ia}^I = 0$ in the TDA case.

C. Differentiable Training Objectives

GradTDDFT treats the fixed molecular data as $\mathcal{I}$ and writes the SCF solution as a fixed point

$$\mathbf{D}_\theta^* = \Phi_\theta(\mathbf{D}_\theta^*; \mathcal{I}), \quad \mathbf{R}_\theta(\mathbf{D}) = \mathbf{D} - \Phi_\theta(\mathbf{D}; \mathcal{I}). \quad (18)$$

The observable map $\mathcal{O}_\theta(\mathbf{D}; \mathcal{I})$ includes ground-state energies, TDDFT excitation energies, and oscillator strengths. A general training loss is written as

$$\mathcal{J}(\theta) = \lambda_{\text{GS}} \mathcal{L}_{\text{GS}} + \lambda_{\text{ES}} \mathcal{L}_{\text{ES}}, \quad (19)$$

where the two terms denote ground-state and excited-state targets. For a given training run in this work, one objective family is selected at a time by setting either $\lambda_{\text{ES}} = 0$ or $\lambda_{\text{GS}} = 0$. This notation therefore describes the two separate protocols within one mathematical framework.

The same forward calculation supports three differentiation choices. In fixed-density training, a reference density is held fixed:

$$\mathcal{J}_{\text{fd}}(\theta) = \ell(\mathcal{O}_\theta(\mathbf{D}^{\text{ref}}; \mathcal{I}), \mathbf{y}), \quad \frac{d\mathbf{D}^{\text{ref}}}{d\theta} = 0. \quad (20)$$

In explicit self-consistent training, a finite SCF trajectory is differentiated directly:

$$\mathbf{D}_{n+1} = \Phi_\theta(\mathbf{D}_n; \mathcal{I}), \quad \mathcal{J}_{\text{exp}}(\theta) = \ell(\mathcal{O}_\theta(\mathbf{D}_N; \mathcal{I}), \mathbf{y}). \quad (21)$$

In implicit self-consistent training, the converged density satisfies $\mathbf{R}_\theta(\mathbf{D}_\theta^*) = 0$, and its parameter derivative follows from

$$\frac{d\mathbf{D}_\theta^*}{d\theta} = - \left(\frac{\partial \mathbf{R}_\theta}{\partial \mathbf{D}} \right)^{-1} \frac{\partial \mathbf{R}_\theta}{\partial \theta}. \quad (22)$$

These equations define the differentiable routes available to the model; the final training schedule can combine them with different weights, targets, and response-kernel constructions.

III. IMPLEMENTATION

The implementation follows the same design principle that underlies GradDFT⁹: fixed electronic-structure inputs are separated from the differentiable program that evaluates energies, self-consistent fields, response matrices, and losses. For GradTDDFT the distinction is extended to excited-state response. Molecular preprocessing constructs the fixed numerical representation of a calculation, while the differentiable core operates on density matrices, orbital rotations, local density variables, and neural functional parameters.

A. Molecular State and Fixed Data

A GradTDDFT calculation begins from the same physical inputs as a conventional KS or TDDFT calculation: nuclear charges and coordinates, charge, spin, basis set, exchange-correlation functional, quadrature grid, and convergence settings. PySCF is used to construct the molecule, basis metadata, quadrature grid, AO values on grid points, one-electron integrals, and electron-repulsion data. These quantities are then treated as fixed arrays during the differentiable SCF and response calculation. This boundary keeps the integral and grid generation path aligned with PySCF reference calculations while placing the density-dependent part of the method inside JAX.

The fixed molecular state contains the overlap matrix S , core Hamiltonian h , grid weights, AO values and derivatives, nuclear repulsion energy, and the selected representation of the two-electron integrals. GradTDDFT supports dense electron-repulsion tensors, density-fitted contractions, and direct J/K construction. The choice of J/K backend changes how Coulomb and exchange contractions are formed, but not the mathematical objects passed to the SCF or TDDFT layers.

B. Differentiable SCF and TDDFT Execution

After molecular preprocessing, the SCF layer is evaluated entirely as a differentiable fixed-point computation. For a density matrix P , the implementation assembles

$$F[P; \theta] = h + J[P] + F_{\text{xc}}[P; \theta] + F_{\text{nl}}[P; \theta], \quad (23)$$

where $J[P]$ is the Coulomb contribution, F_{xc} is the semilocal or neural XC potential, and F_{nl} denotes optional nonlocal HF and MP2/PT2-channel contributions. The SCF map diagonalizes the generalized eigenvalue problem, updates occupations and density, and repeats this operation until the selected convergence criteria are satisfied. The converged state stores orbitals, orbital energies, occupations, density matrices, total energies, and diagnostic residuals needed by both response calculations and training losses.

The differentiable program can be evaluated either through an explicit finite SCF trace or through an implicit fixed-point gradient. Both modes use the same molecular state and Fock-building path. This is important for comparing training objectives: changes in the gradient treatment should not change the forward SCF solution used to build the TDDFT response problem.

The TDDFT layer is differentiated in the same operator-first spirit. For a converged SCF state, the implementation builds matrix actions for the TDA or full Casida response problem from orbital-energy differences, Coulomb and exchange contractions, semilocal XC response actions, and nonlocal neural-channel response terms. During training, the Davidson solver is used only as the forward numerical procedure for obtaining the requested low-lying roots. Following the implicit differentiation formulation for iterative eigensolvers by Lei Wang and co-workers,¹⁸ GradTDDFT uses a differentiable Davidson mode in which the reverse-mode derivative is not taken through the Davidson iteration history, but through the stationarity of the converged Ritz pairs. In practice, the matrix-vector products used inside the Davidson iterations are detached from the reverse pass. After convergence, the Ritz amplitudes are also held fixed and the excitation energy is reconstructed as a Rayleigh quotient of the current response operator. Gradients of excitation-energy losses therefore flow through the response matrix-action builder, the SCF fixed point, and the neural functional parameters, while avoiding a large differentiation tape for the iterative eigensolver. Oscillator strengths and broadened spectra are evaluated from the same forward roots; their present training path does not include an additional eigenvector-response derivative beyond this fixed-amplitude reconstruction. Detailed formulas for the TDA and Casida root-energy derivatives are given in the Supporting Information.

C. Functional Evaluation and Response Core

The functional interface is organized around a local energy-density map. For traditional semilocal functionals, the map evaluates a specified LDA, GGA, or meta-GGA energy density on the grid. For neural functionals, local density descriptors are combined with learned coefficient fields that mix semilocal, HF, and optional MP2/PT2 channels. In both cases the scalar XC energy is obtained by numerical quadrature, and the SCF potential follows from differentiating the same energy expression with respect to the density variables.

The conventional semilocal energy-density components are evaluated through the JAX-XC functional library,¹⁰ with hybrid exact exchange handled by the nonlocal HF channel. JAX-XC provides a broader catalogue of Libxc-translated LDA, GGA, meta-GGA, and hybrid functional routines than the subset used below. For the GradTDDFT implementation and the calculations in this manuscript, the default validated functional inputs include:

- LDA and local-correlation components: LDA exchange, Perdew–Wang correlation, Vosko–Wilk–Nusair correlation, and the VWN-RPA variant.
- GGA components: Becke 88 exchange, PBE exchange, the range-separated PBE exchange component, LYP correlation, and PBE correlation.
- Composite local and semilocal specifications: LDA, SVWN, SVWN-RPA, PBE, and the local semilocal part of LC-WPBE.
- Global and screened hybrid specifications: PBE0/PBEH, B3LYP, BHandHLYP, HSE03, and HSE06. Their semilocal terms are routed through JAX-XC, while the HF fraction is assembled by the exact exchange contraction path.

Additional installed JAX-XC names, including further GGA, meta-GGA, and hybrid components, can be routed as optional channels after validation for the selected SCF and TDDFT response path. They are not part of the default validated set used for the calculations described here.

The TDDFT response layer reuses the converged SCF state and constructs the TDA or full Casida matrices from orbital-energy differences, Coulomb and exchange response contractions, and the XC kernel. For semilocal response, the local kernel is the Hessian of the same energy-density map used for the SCF potential. This gives a single source for E_{xc} ,

Differentiable Response Core for Traditional Functionals

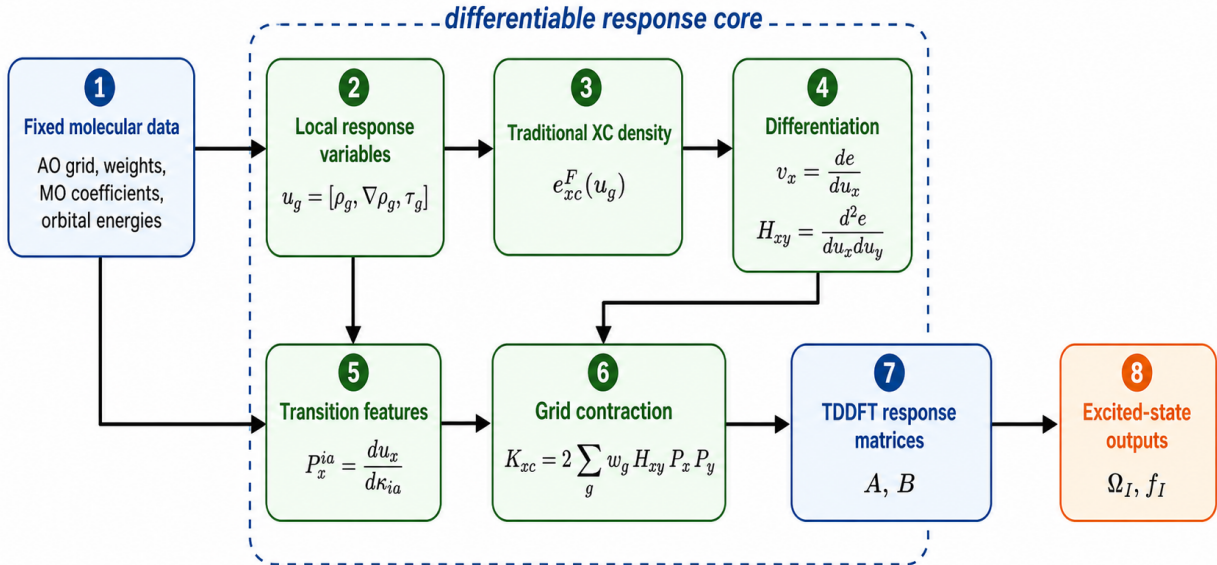


FIG. 1. Differentiable response-core construction for traditional semilocal functionals. Fixed molecular data define local density variables and transition response features, while differentiation of the XC energy density supplies the potential and local Hessian used to assemble TDDFT response matrices.

v_{xc} , and f_{xc} rather than three separately implemented formulas. The corresponding data flow for traditional functionals is summarized in Figure 1.

For the neural HF channel, the response binding converts the learned local exchange field into a density-weighted scalar exchange fraction and inserts it into the same nonlocal exact-exchange contractions used by hybrid TDDFT. For the neural MP2/PT2 channel, the response solver first obtains the TDA or Casida singles roots and then applies a state-specific CIS(D)-type second-order correction scaled by the learned PT2 coefficient. This is the calculation path used for the excited-state tests described in Section IV.

D. Training and Observable Evaluation

The final computational graph maps neural parameters to scalar losses. Starting from the fixed molecular state, GradTDDFT performs SCF, constructs the response matrices, solves the TDA or full TDDFT eigenvalue problem, and evaluates observables such as excitation

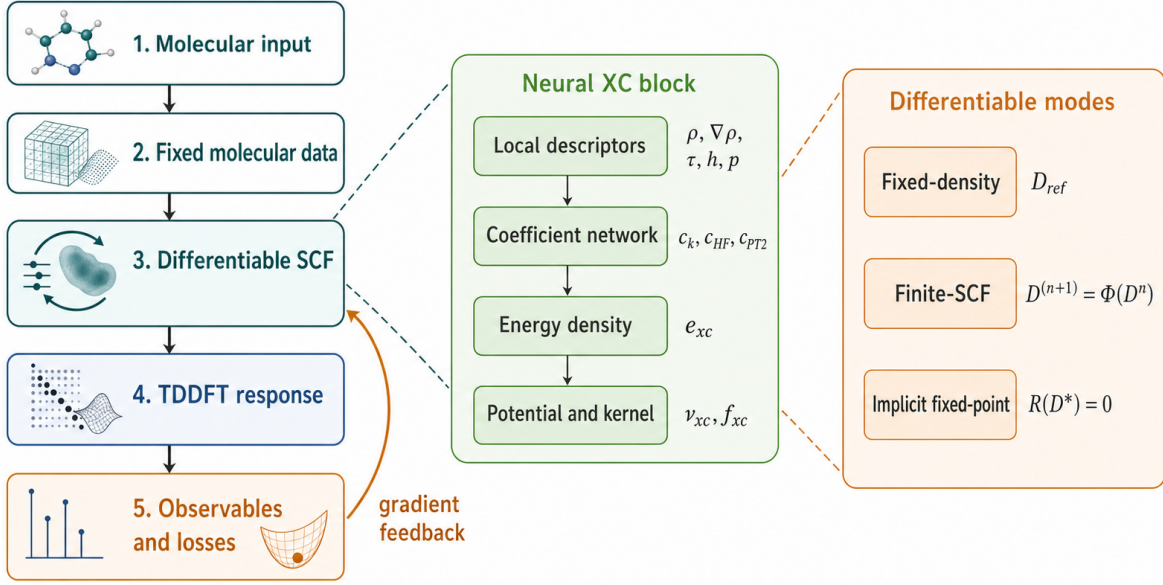


FIG. 2. End-to-end computation flow for differentiable ground-state and excited-state training. Fixed molecular data are passed through SCF and TDDFT response construction to produce observables and scalar losses whose gradients propagate back to the neural functional parameters.

energies, oscillator strengths, transition dipoles, state-ordering metrics. Figure 2 summarizes this end-to-end path.

Loss terms can target ground-state energies, excitation energies, oscillator strengths, or state-ordering diagnostics. The current training protocol optimizes ground-state and excited-state objectives in separate runs. Each calculation records the SCF convergence status, response solver, excited-state treatment, backend, seed, and reference labels. These records separate numerical failures from functional errors.

E. Verification

Verification uses matched PySCF calculations and gradient audits. Runs with traditional functionals compare SCF energies, orbital energies, TDA and full TDDFT excitation energies, oscillator strengths, transition dipoles, and response-matrix components against PySCF references. The matched settings include geometry, basis, grid, functional, and spin treatment. These tests establish that the differentiable path reproduces the reference TDDFT

calculation before neural parameters are introduced.

Differentiability audits evaluate representative scalar losses through the full SCF, response, and observable graph. The audits check finite values, finite gradients, and nonzero parameter sensitivity. They are run for explicit finite-SCF differentiation and implicit fixed-point gradients, together with implicit TDDFT root differentiation, ensuring that the implementation can support the training paradigms described in Section II.

IV. COMPUTATIONAL EVALUATION

The computational evaluation begins with a numerical validation against PySCF and then turns to model-capability tests. The validation module checks that GradTDDFT reproduces conventional TDA and full TDDFT excitation energies, oscillator strengths, and response matrices before neural parameters are introduced. The capability tests then ask whether differentiable neural exchange-correlation functionals can learn ground-state surfaces, excited-state response, and molecular ground-state and vertical-excitation targets within a shared SCF–TDDFT implementation. These model tests are divided into two blocks: bond-dissociation curves covering ground-state surfaces and first excited-state response, and separate ground-state and excited-state training on a 50-molecule QM9 subset.

Across all experiments, the recorded computational state includes the geometry source, charge and spin multiplicity, basis set, quadrature grid, SCF convergence criteria, TDDFT solver, excited-state treatment, backend/device, training seed, and reference method. These records are part of the benchmark definition because small changes in grids, spin treatment, or SCF convergence can otherwise obscure whether an observed error comes from the learned functional or from the numerical setup.

A. Numerical Validation against PySCF

The first evaluation module isolates the TDDFT response construction before any neural functional parameters are trained. In the conventional-functional limit, GradTDDFT evaluates the exchange-correlation energy density as a differentiable function of the density descriptors and obtains both the potential and the local response tensor by automatic

differentiation,

$$E_{\text{xc}} \longrightarrow v_{\text{xc}} = \frac{\delta E_{\text{xc}}}{\delta \rho} \longrightarrow f_{\text{xc}} = \frac{\delta v_{\text{xc}}}{\delta \rho}. \quad (24)$$

The resulting tensor is contracted with quadrature weights and AO basis values, transformed to the occupied–virtual orbital basis, and inserted into the TDA or Casida response matrices. Matched PySCF calculations provide independent references for the same geometry, functional, basis, grid, SCF solution, and response equations.

The validation matrix contains H₂O, CO, N₂, C₂H₄, formaldehyde, benzene, and OH with PBE, B3LYP, and PBE0 in def2-SVP and aug-cc-pVDZ basis sets. Restricted KS references are used for the closed-shell molecules, while unrestricted KS references are used for OH. For each matched setup, TDA and full TDDFT are evaluated for the first five singlet and triplet states where the reference calculation is defined.

For the PBE/def2-SVP subset reported here, all 26 calculations completed successfully, giving 130 per-state records. The direct GradTDDFT–PySCF comparison contains 12 restricted-singlet tasks, corresponding to 60 excitation states across TDA and full TDDFT. The excitation-energy mean absolute differences relative to PySCF range from 5.86×10^{-14} to 2.87×10^{-10} eV, with maximum absolute state errors below 5.47×10^{-10} eV. Oscillator-strength mean absolute differences range from 1.86×10^{-22} to 1.29×10^{-7} . For response matrices small enough to store explicitly in this audit, the RMS differences in the assembled *A* and *B* matrices are 5.65×10^{-15} to 8.26×10^{-15} .

This validation establishes the numerical equivalence of the differentiable TDDFT path and matched PySCF calculations for traditional functionals. The following sections therefore evaluate model behavior rather than basic response-matrix assembly.

B. Ground- and Excited-State Dissociation Curves

The first model-capability benchmark evaluates one-dimensional bond dissociation curves for H₂, H₂⁺, and N₂, with first excited-state scans for H₂ and N₂. The H₂/H₂⁺ pair follows the Kohn–Sham regularizer benchmark of Li *et al.*: H₂ is a compact bond-breaking and strong-correlation test, whereas H₂⁺ isolates one-electron self-interaction and open-shell SCF stability.²² N₂ extends this setting to a stretched multiple bond; recent high-level spectroscopic calculations treat molecular nitrogen as a dense multi-state problem, reporting potential-energy curves and transition dipole moments for the 51 lowest electronic states over

TABLE I. B3LYP/def2-SVP S_1 excitation energies and oscillator strengths from matched PySCF and GradTDDFT calculations. Each entry reports Ω_1 in eV followed by f_1 .

Molecule	Solver	PySCF	GradTDDFT
H ₂ O	TDA	7.6 / 1.8×10^{-2}	7.6 / 1.8×10^{-2}
H ₂ O	TDDFT	7.6 / 1.8×10^{-2}	7.6 / 1.8×10^{-2}
CO	TDA	8.7 / 1.1×10^{-1}	8.7 / 1.1×10^{-1}
CO	TDDFT	8.5 / 8.2×10^{-2}	8.5 / 8.2×10^{-2}
N ₂	TDA	9.5 / 4.5×10^{-31}	9.5 / 3.4×10^{-31}
N ₂	TDDFT	9.4 / 4.7×10^{-18}	9.4 / 7.2×10^{-25}
C ₂ H ₄	TDA	8.3 / 5.4×10^{-29}	8.3 / 6.0×10^{-29}
C ₂ H ₄	TDDFT	8.1 / 3.7×10^{-1}	8.1 / 3.7×10^{-1}
CH ₂ O	TDA	4.0 / 1.9×10^{-29}	4.0 / 1.8×10^{-29}
CH ₂ O	TDDFT	4.0 / 2.2×10^{-25}	4.0 / 5.4×10^{-29}
C ₆ H ₆	TDA	5.5 / 3.7×10^{-9}	5.5 / 3.7×10^{-9}
C ₆ H ₆	TDDFT	5.5 / 4.9×10^{-9}	5.5 / 4.9×10^{-9}

1.35–10 bohr and multiple N + N dissociation limits.²³ The three systems therefore separate self-interaction, two-electron bond breaking, and stretched triple-bond near-degeneracy within a small set of scans. The ground-state curves are labeled by the corresponding diatomic spectroscopic terms, $X^1\Sigma_g^+$ for H₂ and N₂ and $X^2\Sigma_g^+$ for H₂⁺. The excited-state curves are labeled as H₂ $B^1\Sigma_u^+$ and N₂ $A^1\Pi_g$; the latter follows the large-CAS state assignment used in the N₂ reference data.²³

The results are organized around the physically meaningful channel for each system. For the one-electron H₂⁺ ion, the MP2/PT2 doubles space is empty, so a PT2 correction is identically zero and does not provide a useful functional test. H₂⁺ is therefore used to compare implicit-SCF training with fixed-density training. For H₂ and N₂, the implicit-SCF protocol is used throughout, and the PT2/no-PT2 comparison tests how strongly the second-order channel improves the learned functional along bond dissociation. The H₂ and N₂ first excited-state dissociation curves apply the same PT2-channel test to the TDDFT response calculation. The curve-plus-error layout, training markers, and chemical accuracy band are used as direct diagnostics, with convergence success, cycle count, and failure rate

recorded at every geometry.

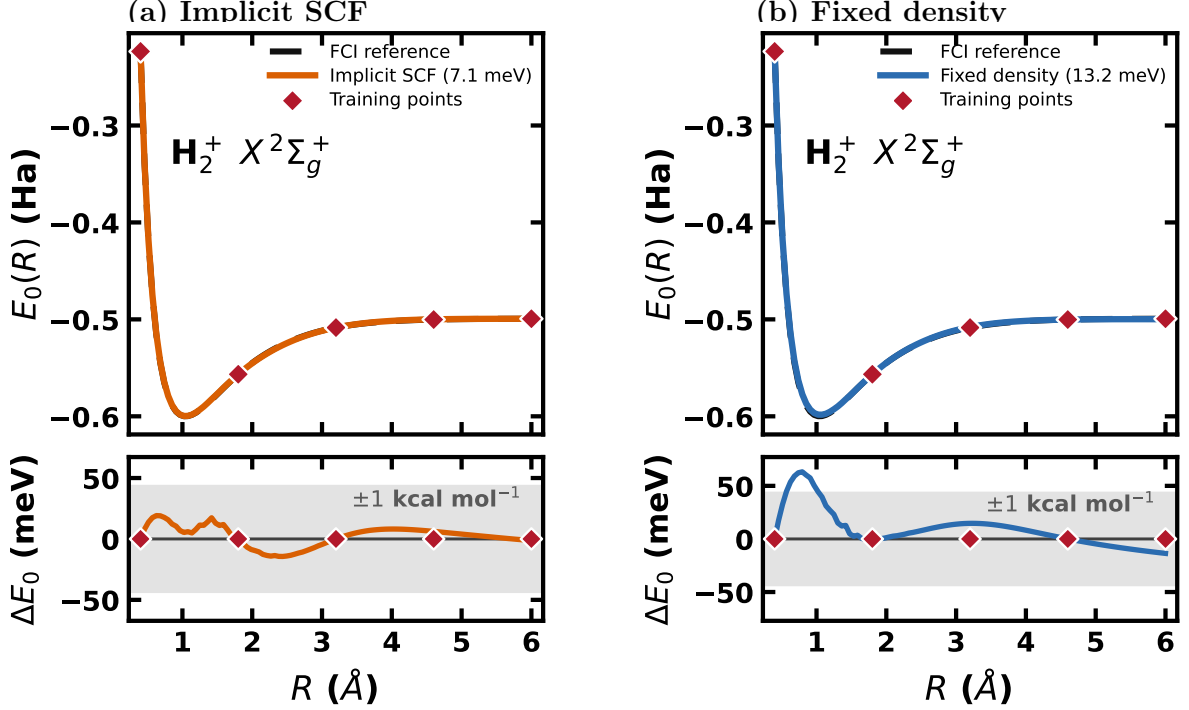


FIG. 3. $\text{H}_2^+ X^2\Sigma_g^+$ ground-state dissociation curves for the HFX-channel neural functional under implicit-SCF and fixed-density training. In each panel, the upper subpanel compares the learned curve with the one-electron FCI reference values, while the lower subpanel shows the signed error $\Delta E_0 = E_{0,\text{model}} - E_{0,\text{ref}}$. Red diamonds denote the five training geometries, and the gray band marks $\pm 1 \text{ kcal mol}^{-1}$. The implicit-SCF model gives a dense-curve mean absolute error of 7.1 meV and a maximum absolute error of 19.2 meV. Fixed-density training gives corresponding errors of 13.2 meV and 63.2 meV.

C. Semi-local Channel Composition Tests

The dissociation benchmarks above vary the implicit-SCF treatment and the second-order PT2 channel. A complementary training test asks a narrower question: under a fixed noPT2+HFX implicit-SCF protocol, how strongly does the semi-local channel basis affect the N_2 ground-state and first-excited-state dissociation curves? In this comparison, the MP2/PT2 channel is disabled, the local exact-exchange descriptor h_g remains active through $c_x^\theta(\mathbf{z}_g)h_g$, and only the set of semi-local energy densities $\{c_m^{\text{sl}}\}$ in Eq. 4 is changed. The neural

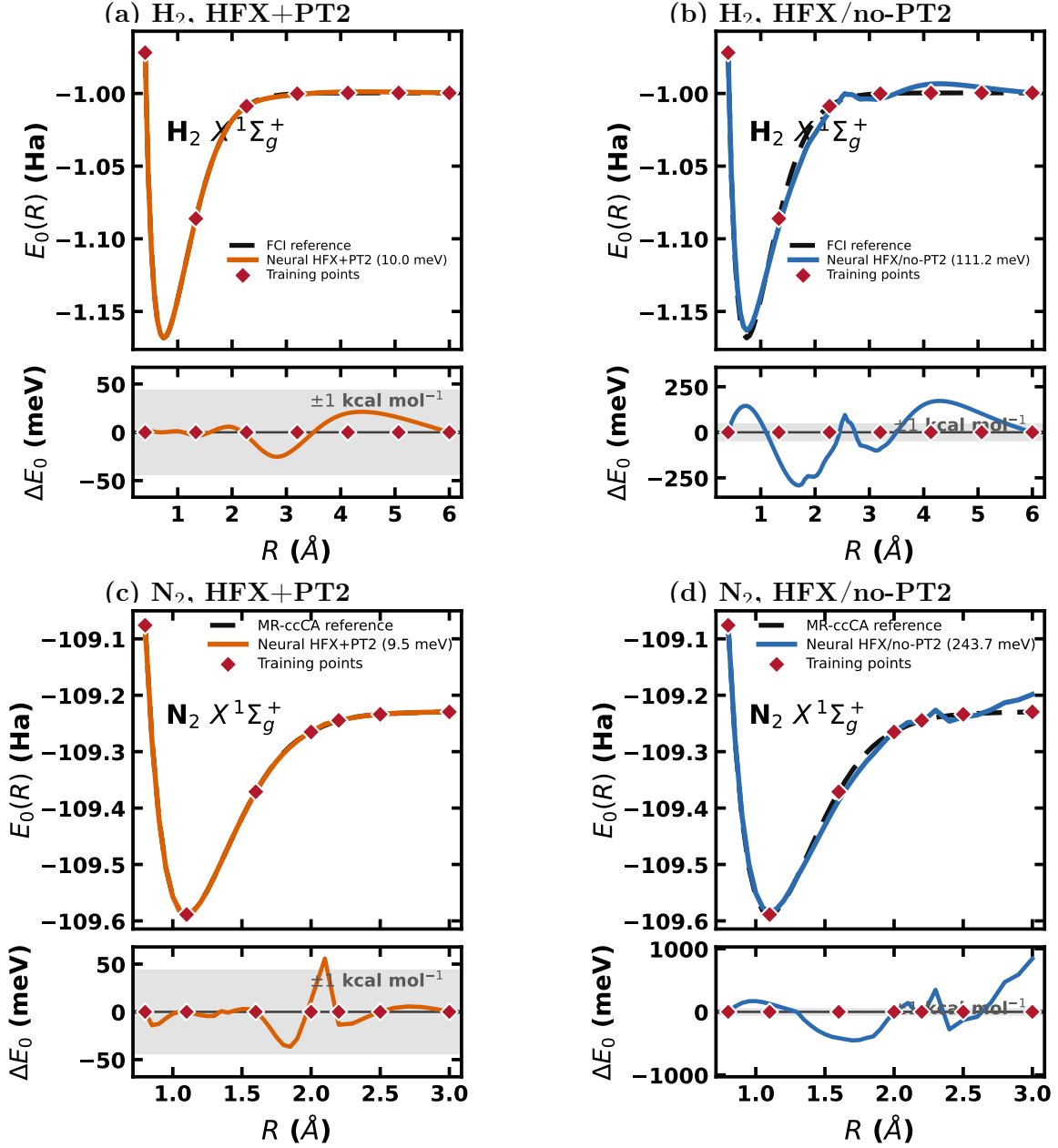


FIG. 4. H_2 and N_2 $X^1\Sigma_g^+$ ground-state dissociation curves for neural functionals trained with and without the PT2 channel. In each panel, the upper subpanel compares the learned curve with the reference values, while the lower subpanel shows the signed error $\Delta E_0 = E_{0,\text{model}} - E_{0,\text{ref}}$. Red diamonds denote the seven training geometries, and the gray band marks ± 1 kcal mol $^{-1}$. For H_2 , the HFX+PT2 model gives a dense-curve mean absolute error of 10.0 meV and a maximum absolute error of 25.4 meV, while removing the PT2 channel increases the corresponding errors to 111.2 meV and 292.1 meV. For N_2 , the HFX+PT2 model gives corresponding errors of 9.5 meV and 55.7 meV, whereas the HFX/no-PT2 model gives 243.7 meV and 844.0 meV.

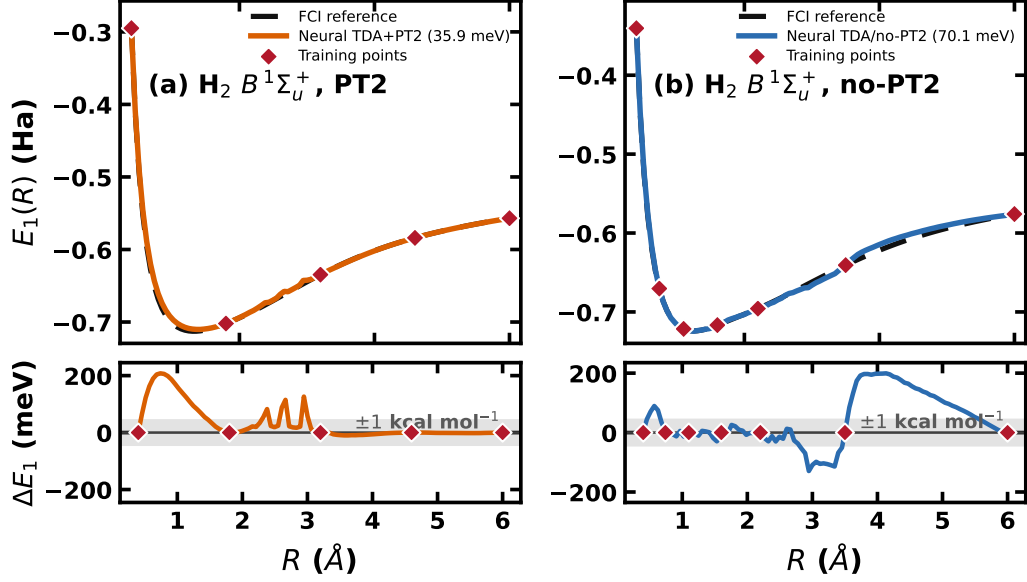


FIG. 5. $\text{H}_2 B^1\Sigma_u^+$ excited-state dissociation curve for neural TDA calculations with and without the PT2 channel. The plotted quantity is the total excited-state energy $E_1(R) = E_0(R) + \Omega_1(R)$. In each panel, the upper subpanel compares the learned curve with the FCI reference values, while the lower subpanel shows the signed error $\Delta E_1 = E_{1,\text{model}} - E_{1,\text{ref}}$. Red diamonds denote the training geometries, and the gray band marks ± 1 kcal mol⁻¹. The PT2-channel model gives a dense-curve mean absolute error of 35.9 meV and a maximum absolute error of 208.0 meV, whereas the updated no-PT2 model gives corresponding errors of 70.1 meV and 199.3 meV.

architecture, optimizer, training geometries, quadrature grid, and implicit differentiation of the SCF-TDDFT workflow are kept fixed.

We refer to these models as semi-local channel-composition variants. The functional labels identify the JAX-XC energy-density components used to build the coefficient basis; they do not introduce fixed parent-functional mixing fractions. Because this benchmark uses the noPT2 setting, no $c_2^\theta(\mathbf{z}_g)p_g$ contribution is included. The comparison therefore isolates how LDA, GGA, and meta-GGA semi-local priors interact with the learned local HFX channel in the coefficient-basis form of Eq. 3.

Each variant is trained and evaluated with the same N_2 geometries, reference labels, basis/grid settings, and response solver. The ground-state comparison uses $E_0(R)$; the excited-state comparison uses the first excitation energy $\Omega_1(R)$ and the total first-excited-state curve $E_1(R) = E_0(R) + \Omega_1(R)$. Since HFX is enabled and PT2 is disabled in all rows of

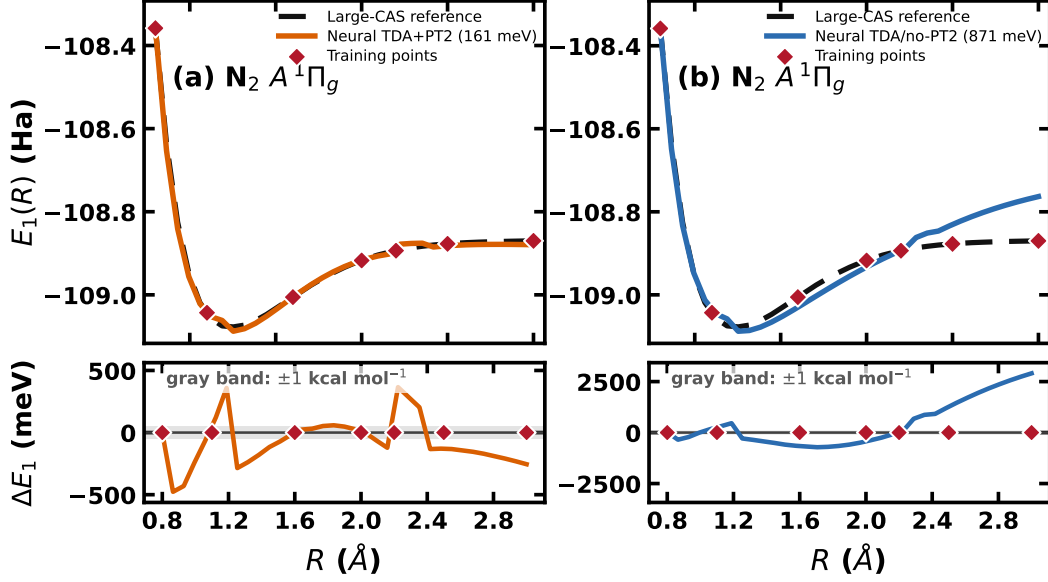


FIG. 6. $\text{N}_2 A^1\Pi_g$ excited-state dissociation curves for neural TDA calculations with and without the PT2 channel. The plotted quantity is the total excited-state energy $E_1(R) = E_0(R) + \Omega_1(R)$, using the Hammami *et al.* large-CAS curve as the reference.²³ In each panel, the upper subpanel compares the learned curve with the reference values, while the lower subpanel shows the signed error $\Delta E_1 = E_{1,\text{model}} - E_{1,\text{ref}}$. Red diamonds denote the training geometries, and the gray band marks $\pm 1 \text{ kcal mol}^{-1}$. The PT2-channel model gives a dense-curve mean absolute error of 0.161 eV and a maximum absolute error of 0.477 eV, while removing the PT2 channel increases the corresponding errors to 0.871 eV and 2.905 eV.

Table II, differences across the three variants are attributed to the semi-local channel basis rather than to changes in the nonlocal response ingredients.

Within the same dissociation benchmark, training on near-equilibrium geometries is compared with training on the full curve. The near-equilibrium regime tests extrapolation into stretched bonds, whereas the full-curve regime tests interpolation across both equilibrium and dissociation regions. This mirrors the GradDFT-style potential-energy-surface logic while adding explicit SCF convergence diagnostics.

The excited-state scans test whether the same differentiable SCF–TDDFT workflow can support response calculations along a dissociation coordinate. For each bond length, the calculation records the ground-state energy $E_0(R)$, the first excitation energy $\Omega_1(R)$, and

TABLE II. Semi-local channel-composition benchmark for N_2 dissociation. The MP2/PT2 channel is disabled, the local HFX channel is enabled, and all variants use the implicit-SCF training path. Only the semi-local basis $\{e_m^{\text{sl}}\}$ is changed.

Variant	Semi-local channel basis	Dissociation diagnostic
lda_vwn_rpa	lda_x + lda_c_vwn_rpa	Baseline for $E_0(R)$ and $E_1(R)$
gga_pbe	gga_x_pbe + gga_c_pbe	GGA exchange/correlation prior
mgga_r2scan	mgga_x_r2scan + mgga_c_r2scan	Meta-GGA τ -dependent prior

the first excited-state energy

$$E_1(R) = E_0(R) + \Omega_1(R). \quad (25)$$

Reference curves are generated with a fixed basis, spin sector, and active space across the scan so that errors reflect the learned functional and response treatment rather than changing reference definitions.

The comparison includes TDA, full TDDFT, and neural-XC TDDFT under the same orbital and response settings. Neural objectives are grouped into ground-state-only and excited-state-only losses. This separation tests whether fitting $E_0(R)$ alone yields a useful response kernel and whether direct S_1 supervision perturbs the ground-state surface. The key diagnostic is the compatibility between the learned SCF fixed point and the learned response kernel along the entire bond scan under the corresponding separate objective.

D. Separate Ground- and Excited-State Training on 50 QM9 Molecules

The second model-capability benchmark evaluates separate ground-state and excited-state training on a 50-molecule subset of QM9.²⁴ The subset contains small organic molecules at fixed ground-state geometries and is used to test whether a neural functional can learn both total-energy and vertical-excitation targets beyond one-dimensional bond scans. The implementation remains self-consistent in both protocols: neural parameters affect the SCF solution, TDDFT response matrices, excitation energies, and oscillator strengths.

The primary targets are ground-state total energies and the lowest vertical excitation energies. Oscillator strengths and state ordering are retained as response-quality diagnostics. Molecule-level splits, random seeds, charge and spin assignments, basis sets, grids, and reference labels are recorded with the training data. Results are reported separately for ground-state-only and excited-state-only objectives, together with molecular holdout errors within the 50-molecule subset, response solver convergence, and SCF failure rate. This block tests whether the differentiable SCF–TDDFT training path can move from controlled dissociation scans to chemically diverse small organic molecules under the two separate training protocols.

V. DISCUSSION

GradTDDFT is designed to make the response kernel part of the trainable computational object rather than an after-the-fact diagnostic. This distinction is important for neural exchange-correlation functionals because excitation energies depend not only on the converged KS orbitals and eigenvalue gaps, but also on the derivative structure of the learned energy density. The energy-consistent local response construction therefore enforces consistency among the neural XC energy, the XC potential, and the local f_{xc} kernel. This consistency is a central requirement when the same functional is used in both SCF optimization and TDDFT response.

The evaluation first separates numerical correctness from model capability. Matched PySCF calculations test whether the differentiable implementation reproduces conventional TDA and full TDDFT in the non-neural limit. After this check, the model-capability benchmarks separate sources of error that are often mixed in neural-functional studies. H_2 , H_2^+ ,

and N_2 ground-state dissociation curves test whether the learned functional remains stable inside an SCF loop across equilibrium and stretched geometries. H_2 and H_2^+ first excited-state dissociation curves then test whether the same functional gives compatible ground-state surfaces and response kernels. The 50-molecule QM9 ground- and excited-state training set tests whether the differentiable SCF–TDDFT training graph transfers from controlled bond scans to chemically diverse small organic molecules under separate total-energy and vertical-excitation objectives. This separation is necessary because a low energy error, a stable SCF solution, and an accurate excitation spectrum are related but not equivalent requirements.

This is also why the present training protocol keeps ground-state and excited-state supervision separate. A joint loss is mathematically straightforward in a differentiable SCF–TDDFT graph, but it is not a robust practical objective for the current neural functional. The ground-state loss primarily drives the self-consistent density, orbital energies, and total-energy surface, whereas the excited-state loss drives the response kernel, excitation energies, oscillator strengths, and state ordering. When both are optimized at the same time, their gradients can compete: updates that improve the ground-state surface may degrade the response kernel, and updates that fit a target excitation can destabilize the SCF fixed point or damage the ground-state energy landscape. In our training design, separating the objectives makes these failure modes observable rather than hiding them inside an unstable multi-target optimization.

The nonlocal channels define a separate theoretical boundary. For semilocal channels, automatic differentiation of the local energy density gives a direct route to a consistent response kernel. For the HF channel, the current excited-state path uses the learned local exchange field to define a scalar hybrid fraction and then applies the ordinary exact-exchange response matrix. For the MP2/PT2 channel, the response problem remains a singles TDA/Casida problem and the learned correlation coefficient scales a root-specific CIS(D)-type doubles correction. This separation is useful for charge-transfer states, double-excitation character, and open-shell systems where the chosen excited-state correction can change the qualitative state ordering.

Several extensions remain natural next steps. A future direct orbital-Hessian treatment of the MP2/PT2 channel would provide a more direct connection between dynamical correlation descriptors and excited-state response. A JAX-native integral path would reduce the separation between the fixed molecular data layer and the differentiable SCF layer,

although efficient production use still requires optimized J/K construction and memory management. General unrestricted differentiable SCF would broaden the training targets to radicals, charged species, and spin-polarized response calculations. Larger basis sets and larger molecules will further test whether the response-kernel construction remains practical when the orbital and quadrature spaces grow.

VI. CONCLUSIONS

We introduced GradTDDFT, a differentiable TDDFT framework for training neural XC functionals with excited-state observables. The framework combines PySCF molecular data with JAX implementations of SCF, functional evaluation, response-kernel construction, and TDA/Casida observables.

GradTDDFT connects neural functional parameters to ground-state energies, excitation energies, and oscillator strengths in one computational graph. This provides a practical platform for testing neural functionals beyond ground-state fitting, including scalar hybrid exchange response and CIS(D)-type PT2 corrections in excited-state calculations.

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SUPPLEMENTARY MATERIAL

The supplementary material contains the following files.

- Detailed theory and response-kernel derivations
- Computational input records for the molecular benchmarks
- Reference geometries, grids, basis choices, and convergence settings used in the reported calculations
- High-resolution versions of the figures in the main text

CODE AVAILABILITY

GradTDDFT is available under the Apache 2.0 license on [GitHub](#). The released code contains the research implementation and supporting materials for this manuscript. Numerical values should be interpreted together with the recorded geometry, basis, grid, convergence, backend, and reference settings.

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